File & Folder Permissions

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Permission is the process of controlling access of a user and group to a certain object.

Permission in Linux are 2 types :

1. Basic Permission
2. Advanced / Special Permission

Basic Permission

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1. chmod : used to control the access of a file and directory
2. chown : used to control the ownership of a file and directory
3. chgrp : used to control the group ownership of a file and directory
4. chage : used to configure password and account policies chmod :-

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Access permission :

1. Read : we can see the content
2. Write : we can edit , modify , delete the content
3. Execute : we can run the file or open the directory

Notation of permissions

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Permission	Symbolic Identification	Numeric Identification
Read	r	4
Write	w	2
Execute	x	1

On whom the permissions are applied

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1. Owner : Who created the file / Directory : **identified by u**
2. Group owner : Group members of the owner : **identified by g**
3. Others : all others who are neither owner nor group members : **identified by o**

Example :

==========

File : doc1.txt

Who created the file → **student —> Owner**

In Student users group there are 3 more members → Teacher1 , Teacher2 , Teacher3

Who are the group members → **teacher1 , teacher2 , teacher3 —> Group owner**

In this system there are some other users as well → Ishan , Rohit , Mouli

Who are others → ishan , rohit , mouli —> They are not a member of student's group also they have not created the doc1.txt file → they are called **OTHERS**

How to see the permission

==========

```
[student1@node2 ~]$
[student1@node2 ~]$ ls -l data/doc1
-rw-r--r--. 1 student1 student1 24 Nov 26 10:14 data/doc1
[student1@node2 ~]$
```

Permission section :

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Initial "-" identifies that the content is a file

rw- : owners permission : owner means who created the file can read and also can edit it

r-- : Group owner permission : the members of owners group can only see the content but can't edit

r-- : Others permission : the other users or group in the system can only see the content but can't edit

chmod use processes / Permission control processes

==========

1. Numeric Method
2. “+/-” Method
3. “=” method

Controlling Permissions using +/- Method

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For Files

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To add write permission to group owner section

==========

```
[student1@node2 ~]$ chmod g+w /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw-rw-r--. 1 student1 student1 76 Nov 26 10:29 /tmp/data/doc1
[student1@node2 ~]$
```

Testing :

```
[student1@node2 ~]$ su - rohit
Password:
Last login: Tue Nov 26 10:30:12 IST 2024 on pts/0
[rohit@node2 ~]$
[rohit@node2 ~]$ vim /tmp/data/doc1
[rohit@node2 ~]$
[rohit@node2 ~]$ cat /tmp/data/doc1
This is a test file
I am student1 & I am the owner & I have edit permission
Ami Rohit Permission peye gechi
yaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaay
[rohit@node2 ~]$
```

How to assign write permission to other section

```
[student1@node2 ~]$ chmod o+w /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw-rw-rw-. 1 student1 student1 148 Nov 26 10:39 /tmp/data/doc1
[student1@node2 ~]$
```

Testing

```
[student1@node2 ~]$ su - mouli
Password:
[mouli@node2 ~]$
[mouli@node2 ~]$ vim /tmp/data/doc1
[mouli@node2 ~]$
[mouli@node2 ~]$ cat /tmp/data/doc1
This is a test file
I am student1 & I am the owner & I have edit permission
Ami Rohit Permission peye gechi
yaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaay
Amader class e akhon tuktuk dhuke distrub korbar chesta korche

[mouli@node2 ~]$
```

Removing write permission from other section

```
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw-rw-rw-. 1 student1 student1 213 Nov 26 10:44 /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ chmod o-w /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw-rw-r--. 1 student1 student1 213 Nov 26 10:44 /tmp/data/doc1
[student1@node2 ~]$
```

Testing

```
[student1@node2 ~]$ su - mouli
Password:
Last login: Tue Nov 26 10:43:54 IST 2024 on pts/0
[mouli@node2 ~]$
[mouli@node2 ~]$ vim /tmp/data/doc1
[mouli@node2 ~]$
```

Removing write permission from group owner section

```
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw-rw-r--. 1 student1 student1 213 Nov 26 10:44 /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ chmod g-w /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw-r--r--. 1 student1 student1 213 Nov 26 10:44 /tmp/data/doc1
[student1@node2 ~]$
```

Removing read permission from group owner

```
[student1@node2 ~]$ chmod g-r /tmp/data/doc1
[student1@node2 ~]$
```

Testing :

```
[student1@node2 ~]$ su - rohit
Password:
Last login: Tue Nov 26 10:50:02 IST 2024 on pts/0
[rohit@node2 ~]$
[rohit@node2 ~]$ vim /tmp/data/doc1
[rohit@node2 ~]$
[rohit@node2 ~]$ cat /tmp/data/doc1
cat: /tmp/data/doc1: Permission denied
[rohit@node2 ~]$
```

Removing read permission from others

```
[student1@node2 ~]$ ls -l /tmp/data/doc1
-rw----r--. 1 student1 student1 213 Nov 26 10:44 /tmp/data/doc1
[student1@node2 ~]$
[student1@node2 ~]$ chmod o-r /tmp/data/doc1
[student1@node2 ~]$
```

Testing

```
[student1@node2 ~]$ su - ishan
Password:
Last login: Tue Nov 26 10:33:04 IST 2024 on pts/0
[ishan@node2 ~]$
[ishan@node2 ~]$ cat /tmp/data/doc1
cat: /tmp/data/doc1: Permission denied
[ishan@node2 ~]$
```

For directory

==========

Default permission for a directory

```
[student1@node2 ~]$ ls -ld /tmp/linux/
drwxr-xr-x. 2 student1 student1 6 Nov 26 11:05 /tmp/linux/
[student1@node2 ~]$
```

Initial d identifies the content is a directory

rwx : permission for owner ; means owner can read the content , write any content , open the content

r-x : permission for group owners : means group members can open and see the content inside it

r-x : permission for group others : means group members can open and see the content inside it

How to add write permission to group owner section

```
[student1@node2 ~]$ chmod g+w /tmp/linux/
[student1@node2 ~]$
[student1@node2 ~]$ ls -ld /tmp/linux/
drwxrwxr-x. 2 student1 student1 19 Nov 26 11:11 /tmp/linux/
```

Testing

```
[student1@node2 ~]$ su - rohit
Password:
Last login: Tue Nov 26 11:12:13 IST 2024 on pts/0
[rohit@node2 ~]$
[rohit@node2 ~]$ cd /tmp/linux/
[rohit@node2 linux]$ touch file2
[rohit@node2 linux]$ ls
file1  file2
[rohit@node2 linux]$ cd
```

How to add write permission to others

```
[student1@node2 ~]$ chmod o+w /tmp/linux
[student1@node2 ~]$
[student1@node2 ~]$ ls -ld /tmp/linux/
drwxrwxrwx. 2 student1 student1 32 Nov 26 11:14 /tmp/linux/
[student1@node2 ~]$
```

Testing

```
[student1@node2 ~]$ su - mouli
Password:
Last login: Tue Nov 26 10:47:33 IST 2024 on pts/0
[mouli@node2 ~]$
[mouli@node2 ~]$ cd /tmp/linux/
[mouli@node2 linux]$ ls
file1  file2
[mouli@node2 linux]$ touch file3
[mouli@node2 linux]$ ls
file1  file2  file3
```

Permission Allocation using “=” method

==========

File1 : -

-rw -r - - r - -

Add write permission to group owner section

chmod g+w file1

-rw -rw - r - -

chmod g=w file1

-rw –w - r - -

Allocate read write permission to group owner section

==========

```
[student@node1 ~]$ chmod g=rw /tmp/database/file1
[student@node1 ~]$
[student@node1 ~]$ ls -l /tmp/database/file1
-rw-rw-r--. 1 student student 12 Nov 28 09:46 /tmp/database/file1
[student@node1 ~]$
```

Allocate read write permission to other section

==========

```
[student@node1 ~]$ chmod o=rw /tmp/database/file1
[student@node1 ~]$
[student@node1 ~]$ ls -l /tmp/database/file1
-rw-rw-rw-. 1 student student 12 Nov 28 09:46 /tmp/database/file1
[student@node1 ~]$
```

Allocating same permission in multiple sections

```
[student@node1 ~]$ chmod ugo=rw /tmp/database/file2
[student@node1 ~]$ ls -l /tmp/database/file2
-rw-rw-rw-. 1 student student 0 Nov 28 09:57 /tmp/database/file2
[student@node1 ~]$
```

Allocating same permission to all sections

```
[student@node1 ~]$ chmod a=r /tmp/database/file2
[student@node1 ~]$ ls -l /tmp/database/file2
-r--r--r--. 1 student student 0 Nov 28 09:57 /tmp/database/file2
[student@node1 ~]$
[student@node1 ~]$ chmod a=wr /tmp/database/file2
[student@node1 ~]$ ls -l /tmp/database/file2
-rw-rw-rw-. 1 student student 0 Nov 28 09:57 /tmp/database/file2
[student@node1 ~]$
```

For directory

==========

Allocating group owner read write execute permission

```
[student@node1 ~]$ chmod g=rwx /tmp/thursday28/
[student@node1 ~]$ ls -ld /tmp/thursday28/
drwxrwxr-x. 2 student student 6 Nov 28 10:02 /tmp/thursday28/
[student@node1 ~]$
```

Allocating others read write execute permission

```
[student@node1 ~]$ chmod o=rwx /tmp/thursday28/
[student@node1 ~]$ ls -ld /tmp/thursday28/
drwxrwxrwx. 2 student student 6 Nov 28 10:02 /tmp/thursday28/
[student@node1 ~]$
```

Allocating group owner and other read write execute permission together

```
[student@node1 ~]$ chmod go=rwx /tmp/thursday28/
[student@node1 ~]$ ls -ld /tmp/thursday28/
drwxrwxrwx. 2 student student 6 Nov 28 10:02 /tmp/thursday28/
[student@node1 ~]$
```

Allocating only read and execute permission to group owner and other

```
[student@node1 ~]$ chmod go=rx /tmp/thursday28/
[student@node1 ~]$ ls -ld /tmp/thursday28/
drwxr-xr-x. 2 student student 6 Nov 28 10:02 /tmp/thursday28/
[student@node1 ~]$
```

Allocating all permission to everyone at a time

```
[student@node1 ~]$ chmod a=rwx /tmp/thursday28/
[student@node1 ~]$ ls -ld /tmp/thursday28/
drwxrwxrwx. 2 student student 6 Nov 28 10:02 /tmp/thursday28/
[student@node1 ~]$
```

If we want to allocate same permission on parent object and we want same permission to be applied on child objects

```
[student@node1 ~]$ chmod a=rwx /tmp/friday29/ -R
[student@node1 ~]$ ls -l /tmp/friday29/
total 0
-rwxrwxrwx. 1 student student 0 Nov 28 10:15 file1
-rwxrwxrwx. 1 student student 0 Nov 28 10:15 file2
[student@node1 ~]$
```

-R represents recursive permission , means if we apply rwx permission on friday29 , -R will apply the rwx permission on files inside the friday29 folder

Numeric Method for permission allocation

==========

Read : 4

Write : 2

Execute : 1

- Section will not be mentioned while allocating permission
- Values of every section must be written while setting permission
- Values are added then allocated to the section
- Means if we want to allocate read and write permission 4+2=6 to be written
- This is the oldest method for allocating permission
- 1st value will denote owner section , 2nd value will denote group section and 3rd value will denote other section
- Sample command : **$ chmod 777 /tmp/file1**
- 1st 7 denotes read+write+execute permission on **owner**
- 2nd 7 denotes read+write+execute permission on **group owner**
- 3rd 7 denotes read+write+execute permission on **others**

Adding write permission to the group owner in the following permission

```
[student@node1 ~]$ ls -l /tmp/sat30/chill
-rw-r--r--. 1 student student 0 Nov 28 10:32 /tmp/sat30/chill
```

Owner section : read + write = 4 + 2 = 6

Group Owner section : read + write = 4 + 2 = 6

Others section : read = 4

Final Permission = 6 6 4

```
[student@node1 ~]$ chmod 664 /tmp/sat30/chill
[student@node1 ~]$ ls -l /tmp/sat30/chill
-rw-rw-r--. 1 student student 0 Nov 28 10:32 /tmp/sat30/chill
[student@node1 ~]$
```

Adding write permission to the other in the following permission

```
[student@node1 ~]$ ls -l /tmp/sat30/chill
-rw-r--r--. 1 student student 0 Nov 28 10:32 /tmp/sat30/chill
```

Owner section : read + write = 4 + 2 = 6

Group Owner section : read = 4

Others section : read + write = 4 + 2 = 6

Final Permission = 6 4 6

```
[student@node1 ~]$ chmod 646 /tmp/sat30/chill
[student@node1 ~]$ ls -l /tmp/sat30/chill
-rw-r--rw-. 1 student student 0 Nov 28 10:32 /tmp/sat30/chill
[student@node1 ~]$
```

To allocate full permission to all sections

Owner = read + write = 4+2 = 6

Group = read + write = 4+2 = 6

Other = read + write = 4+2 = 6

Final Permission = 6 6 6

```
[student@node1 ~]$ chmod 666 /tmp/sat30/chill
[student@node1 ~]$ ls -l /tmp/sat30/chill
-rw-rw-rw-. 1 student student 0 Nov 28 10:32 /tmp/sat30/chill
[student@node1 ~]$
```

For directories

==========

Allocating full permission to a folder to all sections

Owner : read + write + execute = 4 + 2 + 1 = 7

Group : read + write + execute = 4 + 2 + 1 = 7

Other : read + write + execute = 4 + 2 + 1 = 7

```
[student@node1 ~]$ chmod 777 /tmp/sat30/
[student@node1 ~]$ ls -ld /tmp/sat30/
drwxrwxrwx. 2 student student 34 Nov 28 10:32 /tmp/sat30/
[student@node1 ~]$
```

Allocating read write execute permission to group

By default permission : 755

```
[student@node1 ~]$ chmod 775 /tmp/sat30/
[student@node1 ~]$ ls -ld /tmp/sat30/
drwxrwxr-x. 2 student student 34 Nov 28 10:32 /tmp/sat30/
[student@node1 ~]$
```

Allocating read write execute permission to others and group

By default permission : 755

```
[student@node1 ~]$ chmod 777 /tmp/sat30/
[student@node1 ~]$ ls -ld /tmp/sat30/
drwxrwxrwx. 2 student student 34 Nov 28 10:32 /tmp/sat30/
[student@node1 ~]$
```

Verification of recursive feature

```
[student@node1 ~]$ chmod 777 /tmp/sat30/ -R
[student@node1 ~]$ ls -l /tmp/sat30/
total 0
-rwxrwxrwx. 1 student student 0 Nov 28 10:32 chill
-rwxrwxrwx. 1 student student 0 Nov 28 10:32 netflix
[student@node1 ~]$
```

chown

=========

It is used to change ownership of a content.

It can change the owner as well as group owner.

This operation can only be done by root

Changing ownership of a file

```
[root@node1 ~]# ls -l /tmp/sat30/netflix
-rwxrwxrwx. 1 student student 0 Nov 28 10:32 /tmp/sat30/netflix
[root@node1 ~]#
[root@node1 ~]# chown ishan /tmp/sat30/netflix
[root@node1 ~]# ls -l /tmp/sat30/netflix
-rwxrwxrwx. 1 ishan student 0 Nov 28 10:32 /tmp/sat30/netflix
[root@node1 ~]#
```

Changing group ownership of a file

```
[root@node1 ~]# chown :managers /tmp/sat30/netflix
[root@node1 ~]# ls -l /tmp/sat30/netflix
-rwxrwxrwx. 1 ishan managers 0 Nov 28 10:32 /tmp/sat30/netflix
[root@node1 ~]#
```

Changing owner and group owner at a time of a file

```
[root@node1 ~]# chown mouli:admins /tmp/sat30/chill
[root@node1 ~]# ls -l /tmp/sat30/chill
-rwxrwxrwx. 1 mouli admins 0 Nov 28 10:32 /tmp/sat30/chill
[root@node1 ~]#
```

Changing owner and group owner at a time on parent object and flowing it to child objects as well

```
[root@node1 ~]# chown ishan:admins /tmp/sunday1/ -R
[root@node1 ~]# ls -ld /tmp/sunday1/
drwxr-xr-x. 2 ishan admins 42 Nov 28 11:15 /tmp/sunday1/
[root@node1 ~]# ls -l /tmp/sunday1/
total 0
-rw-r--r--. 1 ishan admins 0 Nov 28 11:15 doc1
-rw-r--r--. 1 ishan admins 0 Nov 28 11:15 doc2
-rw-r--r--. 1 ishan admins 0 Nov 28 11:15 doc3
[root@node1 ~]#
```

chgrp :

======

Used to change the group ownership

```
[root@node1 ~]# chgrp managers /tmp/sat30/netflix
[root@node1 ~]# ls -l /tmp/sat30/netflix
-rwxrwxrwx. 1 ishan managers 0 Nov 28 10:32 /tmp/sat30/netflix
[root@node1 ~]#
```

chage :

==========

1. Used to configure account policies and password policy

To see default settings / default password policy

```
[root@node1 ~]# chage -l student
Last password change                                    : Nov 26, 2024
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
[root@node1 ~]#
```

To set minimum password age : how many days we need to wait after account creation for changing the password

```
[root@node1 ~]# chage -m 5 student
[root@node1 ~]# chage -l student
Last password change                                    : Nov 26, 2024
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 5
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
[root@node1 ~]#
```

To set maximum password age : after how many days user must change password

```
[root@node1 ~]# chage -M 25 student
[root@node1 ~]# chage -l student
Last password change                                    : Nov 26, 2024
Password expires                                        : Dec 21, 2024
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 5
Maximum number of days between password change          : 25
Number of days of warning before password expires       : 7
[root@node1 ~]#
```

To change password changing warning age :before how many days user will be prompted to change the password

```
[root@node1 ~]# chage -W 3 student
[root@node1 ~]# chage -l student
Last password change                                    : Nov 26, 2024
Password expires                                        : Dec 21, 2024
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 5
Maximum number of days between password change          : 25
Number of days of warning before password expires       : 3
[root@node1 ~]#
```

To set expiry date of a user

```
[root@node1 ~]# chage -E 2024-12-25 param
[root@node1 ~]# chage -l param
Last password change                                    : Dec 07, 2024
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : Dec 25, 2024
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
[root@node1 ~]#
```

To activate password change at next login

```
[root@node1 ~]# chage -d 0 param
```

```
[student@node1 ~]$ su - param
Password:
You are required to change your password immediately (administrator enforced).
Current password:
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
New password:
Retype new password:
Last login: Sat Dec  7 11:20:05 IST 2024 on pts/0
[param@node1 ~]$
```

Advanced Permission

==========

1. Sticky Bit
2. SGID : set group id
3. SUID : set user id
4. Umask
5. ACL - access control list

ACL : Access Control List

==========

1. It is used to assign permission to specific users and groups.

2. Permissions are for others remains same

Apply Read & Write Permission on only morium user

```
[root@node2 ~]# setfacl -m u:morium:rw /tmp/tuesday3/doc1
[root@node2 ~]#
```

Verification

```
[root@node2 ~]# getfacl /tmp/tuesday3/doc1
getfacl: Removing leading '/' from absolute path names
# file: tmp/tuesday3/doc1
# owner: student
# group: student
user::rw-
user:morium:rw-
group::r--
mask::rw-
other::r--
```

To remove all permission from jeet user

```
[root@node2 ~]# setfacl -m u:jeet:--- /tmp/tuesday3/doc1
```

To apply read write execute permission for a group named admins

Admins group members are - ishan and biraj

```
[root@node2 ~]# setfacl -m g:admins:rwx /tmp/tuesday3/doc1
[root@node2 ~]#
```

To remove acl from morium user

```
[root@node2 ~]# setfacl -x u:morium /tmp/tuesday3/doc1
[root@node2 ~]#
```

To remove acl from a group admins

```
[root@node2 ~]# setfacl -x g:admins /tmp/tuesday3/doc1
[root@node2 ~]#
```

Sticky Bit :-

1. It is a special permission
2. It is applied on others section
3. It is used to restrict edit and delete of contents created by other users in a public location
4. Symbolic representation of sticky bit is “t”
5. Numeric representation of sticky bit is 1
6. It is by default applied on **/tmp**

7. Numerically it is situated before normal permissions.

Lab :

Applying sticky bit symbolically

```
[root@node1 ~]# chmod o+t /database/
```

Applying sticky bit numerically

```
[root@node1 ~]# chmod 1777 /database/
[root@node1 ~]#
```

Task :

1. create 3 users - user1 , user2 , user3 - password 123
2. Create a directory from root named /data
3. Now apply full permission
4. Now apply sticky bit
5. Test it from any 2 user

umask :

=========

1. When we create any file or directory in linux , these contents get automatic permissions.

2. umask assigns the default permission to the contents.
3. It is a 4 character value.
4. Default value is 0022
    a. 0 : denotes special permission
    b. 0 : permission for owner
    c. 2 : permission for group
    d. 2 : permission for others.

```
How umask is used to assign permissions
=========================================
for files :
============
full permission : 0  6  6  6
umask           : 0  0  2  2
-----------------------------------
default perm    : 0  6  4  4
-----------------------------------
0 : special permission
6 : read write (4+2) for owner
4 : read only for group owner
4 : read only for others
```

```
for directories
=================
Full Permission : 0  7  7  7
umask           : 0  0  2  2
=================================
default Perm    : 0  7  5  5
=================================
0 : special permission
7 : read + write + execute = 4+2+1 = for owner
5 : read + execute = 4+1 = for group owner
5 : read + execute = 4+1 = for others
```

Changing umask value

==========

- Temporary change : valid till logout , when user will login again default value will be restored.
- Permanent Change : valid until changed

How to change umask value : Temporary

```
[student@node1 ~]$ umask 0222
[student@node1 ~]$
[student@node1 ~]$ mkdir test1
[student@node1 ~]$ ls -ld test1
dr-xr-xr-x. 2 student student 6 Dec  7 10:06 test1
[student@node1 ~]$
```

Creating a custom logout environment for a user

==========

1. vim ~/.bash_logout
2. Type the commands to be executed
3. Save and exit

Alias

======

```
[student@node1 ~]$ alias khoka='mkdir -p'
[student@node1 ~]$
[student@node1 ~]$ khoka rohit
[student@node1 ~]$ ls
a  database  exam  kk373  rohit  test1  weekend
[student@node1 ~]$
```

```
[student@node1 ~]$ alias ishan='ls -l'
[student@node1 ~]$
[student@node1 ~]$ ishan database/
total 8
-rw-r--r--. 1 student student 115 Nov 28 09:04 file1
-rw-r--r--. 1 student student  67 Nov 28 09:14 file2
drwxr-xr-x. 2 student student   6 Nov 28 08:48 mysql
[student@node1 ~]$
```

How to reset root password

==========

1. Restart the system
2. Interrupt the startup using up/down arrow key
3. Select rescue kernel
4. Press e
5. Go to the line starting with linux
6. Press ctrl+E to go to the last of the line
7. Type rd.break then ctrl+x
8. mount -o remount,rw /sysroot
9. chroot /sysroot
10. passwd root
11. Set password
12. Confirm password
13. Update timestamp : touch /.autorelabel
14. exit
15. exit